# Phase 1: Corpus Summary Report

Team: [names here] | Course: Information Retrieval

## 1. Corpus construction

The corpus has 550 papers from arXiv category cs.CL, submitted between 202301010000 and 202412312359 (YYYYMMDDHHMM). Papers were collected with the arxiv Python package, one search per sub-topic, sorted by relevance. PDFs were downloaded and the text was extracted with PyMuPDF.

Exact queries used:

```
machine translation: cat:cs.CL AND abs:"machine translation" AND submittedDate:[202301010000 TO 202412312359]
question answering: cat:cs.CL AND abs:"question answering" AND submittedDate:[202301010000 TO 202412312359]
summarization: cat:cs.CL AND abs:"summarization" AND submittedDate:[202301010000 TO 202412312359]
dialogue systems: cat:cs.CL AND abs:"dialogue system" AND submittedDate:[202301010000 TO 202412312359]
sentiment analysis: cat:cs.CL AND abs:"sentiment analysis" AND submittedDate:[202301010000 TO 202412312359]
```

## 2. Sub-topic coverage

| subtopic | papers |
|---|---|
| machine translation | 110 |
| question answering | 110 |
| summarization | 110 |
| dialogue systems | 110 |
| sentiment analysis | 110 |

## 3. Length variants

Short = 6 pages or fewer, medium = 7 to 9 pages, long = more than 9 pages. The assignment leaves 6-page papers undefined, so we counted them as short.

| variant | papers |
|---|---|
| long | 390 |
| medium | 84 |
| short | 76 |

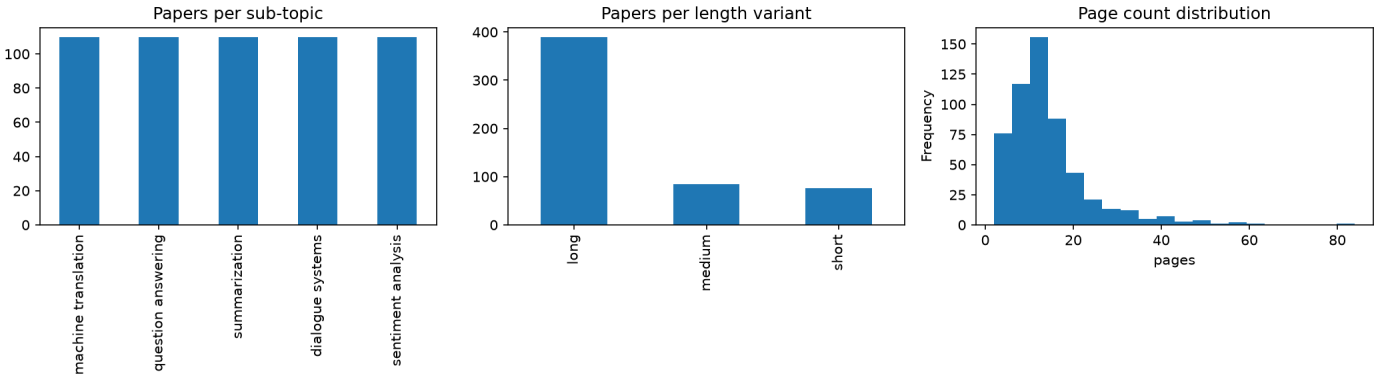
Sub-topic by length variant:

| subtopic | long | medium | short |
|---|---|---|---|
| dialogue systems | 77 | 18 | 15 |
| machine translation | 81 | 15 | 14 |
| question answering | 78 | 16 | 16 |
| sentiment analysis | 68 | 24 | 18 |

| | | | |
|---|---|---|---|
| summarization | 86 | 11 | 13 |

## 4. Document statistics

Pages per paper: mean 14.6, min 2.0, max 84.0. Words per paper: mean 7990, min 1426, max 36184.




## 5. Stored data

Each record in data/corpus.json holds: arxiv_id, title, link, subtopic, n_pages, variant and the full extracted text. The original PDFs are in data/pdfs/. Rerunning the retrieval notebook with the queries above reproduces the corpus.

## 6. Limitations

Text was extracted from PDFs, so it includes headers, references and some formatting noise. Papers matching more than one sub-topic keep only the first label. The dense retriever is LSA (TF-IDF + truncated SVD) because the neural model could not be downloaded.

## 7. AI usage disclosure

AI assistance (Claude) was used while building the code for this phase. Chat transcripts are in /ai-logs/phase1/.